

Classical Mechanics

Midterm Exam

November 11, 2022

Question 1 (1 point)

A meson π^+ is created and travels a distance $d = 780\text{m}$ at constant velocity $v = 0.99995c$ before it disintegrates. How long does it take for the meson to disintegrate as measured in its own reference frame?

Question 2 (1 point)

Consider a particle of mass m moving in one dimension under a potential $V(x)$. Prove that the Euler-Lagrange equation for the particle is equivalent to Newton's second law.

Question 3 (2 points)

Consider the 1-dimensional wave equation given by

$$\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi(x, t) = \frac{\partial^2}{\partial x^2} \phi(x, t) .$$

Prove that this equation is invariant under a Lorentz boost in the x direction, assuming that $\phi(x, t)$ is an invariant scalar function $\phi(x, t) = \phi'(x', t')$.

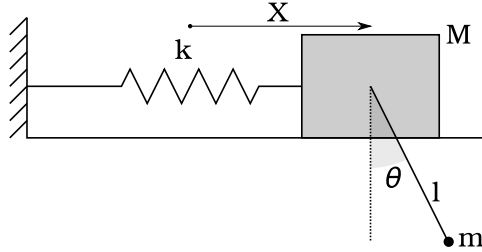
Exercise 1 (3 points)

A stationary mass M_A decays into masses M_B and M_C .

- Calculate the energies of M_B and M_C as a function of the three masses, in the same reference frame where M_A was stationary.
- Calculate the momenta of M_B and M_C as a function of the three masses, in the same reference frame where M_A was stationary. (**Hint:** The resulting expressions are not particularly nice, so do not worry about beautifying them!)
- Verify that if $M_B + M_C = M_A$, then $E_B = M_B$ and $E_C = M_C$. What is the momentum in this case? What does this mean and why does this happen?

Exercise 2 (3 points)

A block with mass M is connected to the wall by a spring with constant k ; the block can slide without friction on a horizontal rail. Additionally, a rigid pendulum of mass m and fixed length ℓ is suspended (subject to gravity) from the center of the block. The whole system is shown in the following figure:



- Write the Lagrangian of the system in terms of the horizontal position X of the block, and the angle θ of the pendulum with respect to the vertical direction.
- Derive the equations of motion for X and θ .
- Obtain the trajectory $X(t)$ of the block in the limit $k = M \gg m$; you can use the initial conditions $X(t = 0) = X_0$ and $\dot{X}(t = 0) = 0$ or leave the solution in terms of arbitrary integration constants. Then, obtain the equation of motion for θ in this limit (using the explicit expression of the integrated $X(t)$ into the equation of motion for θ).

(Q1)

τ : duration in Earth reference frame

τ' : " " π^+ " " "

$$\tau = \frac{d}{v} ; \quad \tau' = \frac{\tau}{\gamma} = \frac{d}{v} \sqrt{1 - \frac{v^2}{c^2}} = 2.6 \cdot 10^{-8} \text{ s}$$

$0.99995 c$ time dilation (shortest time is in the π^+ reference frame: longer in any other frame)

(Q2)

$$\mathcal{L} = \frac{m}{2} \dot{x}^2 - V(x)$$

$$\left. \begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) &= \frac{d}{dt} (m \dot{x}) = m \ddot{x} \\ \frac{\partial \mathcal{L}}{\partial x} &= - \frac{dV}{dx} \end{aligned} \right\} \xrightarrow{E-L} m \ddot{x} = - \underbrace{\frac{dV}{dx}}_{\text{force}} \quad \square$$

Q3

$$\frac{1}{c^2} \frac{\partial^2 \phi(x, t)}{\partial t^2} = \frac{\partial^2 \phi(x, t)}{\partial x^2}$$

$$\begin{aligned} x' &= \gamma(x - vt) & \frac{\partial x'}{\partial x} &= \gamma & ; & \frac{\partial x'}{\partial t} = -\gamma v \\ t' &= \gamma(t - \frac{v}{c^2}x) & \frac{\partial t'}{\partial x} &= -\frac{v}{c^2}\gamma & , & \frac{\partial t'}{\partial t} = \gamma \end{aligned}$$

$$(X) \quad \frac{\partial}{\partial x} = \frac{\partial x'}{\partial x} \frac{\partial}{\partial x'} + \frac{\partial t'}{\partial x} \frac{\partial}{\partial t'} = \gamma \frac{\partial}{\partial x'} - \frac{v}{c^2} \gamma \frac{\partial}{\partial t'}$$

$$\begin{aligned} \frac{\partial^2}{\partial x^2} &= \frac{\partial x'}{\partial x} \frac{\partial}{\partial x'} \left(\gamma \frac{\partial}{\partial x'} - \frac{v}{c^2} \gamma \frac{\partial}{\partial t'} \right) + \frac{\partial t'}{\partial x} \frac{\partial}{\partial t'} \left(\gamma \frac{\partial}{\partial x'} - \frac{v}{c^2} \gamma \frac{\partial}{\partial t'} \right) \\ &= \gamma^2 \frac{\partial^2}{\partial x'^2} - \frac{v}{c^2} \gamma^2 \frac{\partial^2}{\partial x' \partial t'} - \frac{v}{c^2} \gamma^2 \frac{\partial^2}{\partial t' \partial x'} + \frac{v^2}{c^4} \gamma^2 \frac{\partial^2}{\partial t'^2} \end{aligned}$$

$$(t) \quad \frac{\partial}{\partial t} = \frac{\partial x'}{\partial t} \frac{\partial}{\partial x'} + \frac{\partial t'}{\partial t} \frac{\partial}{\partial t'} = -\gamma v \frac{\partial}{\partial x'} + \gamma \frac{\partial}{\partial t'}$$

$$\begin{aligned} \frac{\partial^2}{\partial t^2} &= \frac{\partial x'}{\partial t} \frac{\partial}{\partial x'} \left(-\gamma v \frac{\partial}{\partial x'} + \gamma \frac{\partial}{\partial t'} \right) + \frac{\partial t'}{\partial t} \frac{\partial}{\partial t'} \left(-\gamma v \frac{\partial}{\partial x'} + \gamma \frac{\partial}{\partial t'} \right) \\ &= \gamma^2 v^2 \frac{\partial^2}{\partial x'^2} - \gamma^2 v \frac{\partial^2}{\partial x' \partial t'} - \gamma^2 v \frac{\partial^2}{\partial x' \partial t'} + \gamma^2 \frac{\partial^2}{\partial t'^2} \end{aligned}$$

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = \frac{\partial^2 \phi}{\partial x'^2} \Rightarrow \gamma^2 \frac{v^2}{c^2} \frac{\partial^2}{\partial x'^2} - \cancel{2\gamma^2 v \frac{\partial^2}{\partial x' \partial t'}} + \gamma^2 \frac{\partial^2}{\partial t'^2} =$$

$$= \gamma^2 \frac{\partial^2}{\partial x'^2} - \cancel{\frac{2v}{c^2} \gamma^2 \frac{\partial^2}{\partial x' \partial t'}} + \frac{v^2}{c^4} \gamma^2 \frac{\partial^2}{\partial t'^2} \Rightarrow \phi = \phi' \text{ invariant}$$

$$\Rightarrow \frac{\gamma^2}{c^2} \left(1 - \frac{v^2}{c^2} \right) \frac{\partial^2 \phi}{\partial t'^2} = \gamma^2 \left(1 - \frac{v^2}{c^2} \right) \frac{\partial^2 \phi}{\partial x'^2} \Rightarrow \boxed{\frac{1}{c^2} \frac{\partial^2 \phi'}{\partial t'^2} = \frac{\partial^2 \phi'}{\partial x'^2}}$$

Ex 1

Before
 $M_A, P_A = 0$

After

M_B, M_C

$$P^2 = -E^2 + \vec{P}^2 = -M^2$$

$$\Rightarrow E^2 = M^2 + \vec{P}^2$$

$$\vec{P}^2 = E^2 - M^2$$

Before

$$P_A = (M_A, 0)$$

After

$$P_B = (E_B, \sqrt{E_B^2 - M_B^2})$$

$$P_C = (M_A - E_B, -\sqrt{E_B^2 - M_B^2})$$

$$M_C^2 = (M_A - E_B)^2 - (E_B^2 - M_B^2) = M_A^2 + \cancel{E_B^2} - 2M_A E_B - \cancel{E_B^2} + M_B^2$$

$$\Rightarrow E_B = \frac{M_A^2 + M_B^2 - M_C^2}{2M_A}$$

$$E_C = M_A - E_B = \frac{2M_A^2 - (M_A^2 + M_B^2 - M_C^2)}{2M_A} =$$

$$= \frac{M_A^2 + M_C^2 - M_B^2}{2M_A}$$

$$P = \sqrt{E_B^2 - M_B^2} = \frac{1}{2M_A} \sqrt{M_A^4 + M_B^4 + M_C^4 + 2M_A^2 M_B^2 - 2M_A^2 M_C^2 - 2M_B^2 M_C^2 - 4M_A^2 M_B^2} =$$

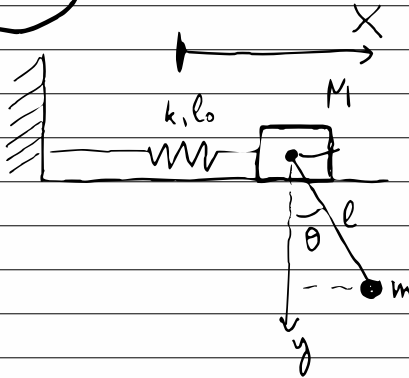
$$= \frac{1}{2M_A} \sqrt{M_A^4 + M_B^4 + M_C^4 - 2M_A^2 M_B^2 - 2M_A^2 M_C^2 - 2M_B^2 M_C^2}$$

$$b) M_A = M_B + M_C \Rightarrow E_B = \frac{M_B^2 + M_C^2 + 2M_B M_C + M_B^2 - M_C^2}{2M_A} = \frac{2M_B(M_B + M_C)}{2M_A}$$

$$= M_B \Rightarrow P = 0$$

Ex 2

a)



a) $X \rightarrow \dot{X}^2$

$$x = X + l \sin \theta$$

$$y = l \cos \theta$$

$$\dot{x} = \dot{X} + l \cos \theta \dot{\theta} ; \dot{x}^2 = \dot{X}^2 + l^2 \cos^2 \theta \dot{\theta}^2 + 2 \dot{X} \dot{\theta} l \cos \theta$$

$$\dot{y} = -l \sin \theta \dot{\theta} ; \dot{y}^2 = l^2 \sin^2 \theta \dot{\theta}^2$$

$$\dot{x}^2 + \dot{y}^2 = \dot{X}^2 + l^2 \dot{\theta}^2 + 2 l \dot{X} \dot{\theta} \cos \theta$$

$$\mathcal{L} = \frac{M}{2} \dot{X}^2 + \frac{m}{2} (\dot{X}^2 + l^2 \dot{\theta}^2 + 2 \dot{X} \dot{\theta} l \cos \theta) - \frac{k}{2} X^2 + m g l \cos \theta$$

$$= \frac{M+m}{2} \dot{X}^2 + \frac{m}{2} l^2 \dot{\theta}^2 + m l \dot{X} \dot{\theta} \cos \theta - \frac{k}{2} X^2 + m g l \cos \theta$$

$$b) \boxed{X} \quad \frac{\partial \mathcal{L}}{\partial X} = -kX$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{X}} \right) = \frac{d}{dt} \left((M+m) \dot{X} + m l \dot{\theta} \cos \theta \right) = (M+m) \ddot{X} + m l \cos \theta \ddot{\theta} - \left. \begin{aligned} & - m l \dot{\theta}^2 \sin \theta \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow \boxed{(M+m) \ddot{X} + m l (\cos \theta \ddot{\theta} - \dot{\theta}^2 \sin \theta) = -kX}$$

$$\boxed{\theta} \quad \frac{\partial \mathcal{L}}{\partial \theta} = -m l \dot{X} \dot{\theta} \sin \theta - m g l \sin \theta$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = \frac{d}{dt} \left(m l^2 \ddot{\theta} + m l \ddot{X} \cos \theta \right) = m l^2 \ddot{\theta} + m l \ddot{X} \cos \theta \left. \begin{aligned} & - m l \dot{X} \sin \theta \dot{\theta} \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow m l^2 \ddot{\theta} + m l \ddot{X} \cos \theta - \cancel{m l \dot{X} \sin \theta \dot{\theta}} = - \cancel{m l \dot{X} \sin \theta \dot{\theta}} - m g l \sin \theta$$

$$\Rightarrow \boxed{\cancel{m l^2} \ddot{\theta} + \cancel{m l} \ddot{X} \cos \theta = - \cancel{m g l} \sin \theta} \Rightarrow$$

$$\Rightarrow \boxed{l \ddot{\theta} = - \ddot{X} \cos \theta - g \sin \theta}$$